Maximality of the Došen–Petrić Boolean Category

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Abstract

We prove that the zero-identity Boolean category of Došen and Petrić freely generated by a nonempty set of propositional variables is maximal. Adjoining an equation between distinct parallel arrows and closing under the arrow operations and uniform formula substitution identifies all parallel arrows. This disproves the conjecture in Section 14.2 of the September 2007 revision of *Proof-Theoretical Coherence*. Boolean Coherence and composition with arrows selecting occurrences reduce the added equation to $\iota_1^{\ell,\top} \circ \pi_1^{\ell,\bot} = 0_{\ell\wedge\bot,\ell\vee\top}$ for a literal ℓ. Uniform substitution and double-negation isomorphisms give this equation with any formula A in place of ℓ. Two isomorphisms listed by Došen and Petrić then yield equality of the product projections. The Lean formalization proves coherence, maximality and the model corollary for the category presented by the defining equations. It also proves the equations relating substitution on tagged occurrence sets to substitution on finite ordinals.

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1 Introduction

Došen and Petrić conjecture that their freely generated zero-identity Boolean category is not maximal [1, §14.2, p. 322]. We prove that it is maximal. Any equation between parallel arrows that does not already hold, when closed under the arrow operations and uniform formula substitution, identifies every pair of parallel arrows. The category before adjoining the equation is not a preorder: for any propositional variable p, its two projections $p \wedge p \to p$ are distinct.

By Boolean Coherence, distinct parallel arrows have different relations between occurrences of propositional variables. We select a pair belonging to one relation and not the other. Precomposition and postcomposition isolate this pair and give the equation used in Došen and Petrić's relative-maximality argument [1, §9.7, pp. 215–216]. Two isomorphisms listed in [1, §12.5, p. 294] then give equality of the product projections. The proof of [2, Lemma 7] also supplies these isomorphisms.

For cartesian categories, Došen and Petrić prove maximality by using coherence, substitution and composition to derive equality of the product projections [3, §5]. For cartesian closed categories, they obtain the same conclusion from a typed Böhm theorem [4, §11]. The latter result does not apply to B: the exponential $p \Rightarrow p$ does not exist, as shown in Section 3. Sections 4 and 5 give the reduction to equal projections in B.

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Došen and Petrić prove maximality for lattice categories and relative maximality for dicartesian categories [5, §§8–10], without assuming distribution. Their book proves maximality for zero-identity distributive lattice categories without the truth constants, and relative maximality when both constants are present [1, §12.5, p. 292]. For formulas A, B containing no negation, with $A \to B$ a tautology, [2, Lemma 6] constructs an arrow $A \to B$ whose occurrence relation consists of the edge between any prescribed occurrences of the same variable in A and B. In [6, §16], they isolate a pair belonging to one relation and not the other to prove maximality of the presentations of split preorders, split equivalence relations and binary relations.

For multiplicative linear logic without constants, Došen and Petrić prove coherence for proof-net categories, with and without mix, with respect to Brauerian split equivalences [7].

Other definitions of Boolean category do not require conjunction and disjunction to be categorical products and coproducts. Lamarche and Straßburger construct free categories from classical proof nets for several systems of equations on the logical operations [8, Theorem 3.5.4]. Straßburger studies equations on monoid and comonoid structures in $*$-autonomous categories [9]. Führmann and Pym interpret cut reduction by inequalities between arrows. An inequality belongs to a net theory if and only if it holds in every interpretation of that theory in their classical categories [10, Theorems 6.4–6.5]. The negation equations differ from those used here. With their monoidal unit isomorphisms, the composite in (2) equals 1_A by their negation triangle [10, (11)]; in Došen and Petrić's category it is 0_A.

We use the occurrence coordinates of [11, §9]: occurrences in a compound formula form tagged disjoint unions, and substitution replaces each occurrence by its own block of occurrences. Section 3 describes these blocks, and Appendix A gives the commutative diagrams relating them to Došen and Petrić's ordinal substitution operation. Section 6 states the Lean theorems and their hypotheses. Appendix C records two typographical corrections to the cut-elimination proof in [1]. References to [1] use the printed pagination of its September 2007 revised version.

2 The Boolean proof category

Fix a nonempty set P of propositional variables. Write $\mathrm{Form}(P)$ for the formulas built from P, the constants $\top, \bot$, the binary connectives $\wedge, \vee$, and negation $\neg$. Let $\mathsf{B}(P)$ be the freely generated zero-identity Boolean category defined in [1, §§14.2–14.3]. Its objects are these formulas, and its arrows are the arrow terms of that presentation modulo its equations. We write B when P is fixed. We recall the structure and equations used in the proof.

The connectives $\wedge$ and $\vee$ give binary products and coproducts; $\top$ is terminal and $\bot$ is initial. For $A_1 = A$ and $A_2 = B$, denote the projections and injections by

$$\pi_i^{A,B} : A \wedge B \longrightarrow A_i, \qquad \iota_i^{A,B} : A_i \longrightarrow A \vee B \qquad (i = 1, 2).$$

For $f : X \to A$ and $g : X \to B$, write $\langle f, g \rangle : X \to A \wedge B$ for their pairing. For $f : A \to X$ and $g : B \to X$, write $[f, g] : A \vee B \to X$ for their copairing. Write $!_A : A \to \top$ and $\kappa_A : \bot \to A$ for the terminal and initial arrows, and $1_A : A \to A$ for the identity. Composition $g \circ f$ applies f first. For $f : A \to A'$ and $g : C \to C'$, the product and coproduct bifunctors give arrows

$$f \wedge g : A \wedge C \longrightarrow A' \wedge C', \qquad f \vee g : A \vee C \longrightarrow A' \vee C'.$$

The dissociativity arrow is

$$d_{A,B,C} : A \wedge (B \vee C) \longrightarrow (A \wedge B) \vee C.$$

Its first-injection equation is

$$d_{A,B,C} \circ (1_A \wedge \iota_1^{B,C}) = \iota_1^{A \wedge B, C}. \tag{1}$$

This is the coproduct-injection equation of [1, §11.1, p. 234].

Negation is represented by a functor $\neg : \mathsf{B}^{\mathrm{op}} \to \mathsf{B}$, sending $f : A \to C$ to $\neg f : \neg C \to \neg A$. Its double-negation and De Morgan isomorphisms are natural. In particular, write $n_A : \neg\neg A \to A$ for the double-negation isomorphism. The presentation also has arrows $\eta_A : \top \to \neg A \vee A$ and $\epsilon_A : A \wedge \neg A \to \bot$. The *zero identity* is the endomorphism $0_A : A \to A$ defined by

$$0_A = [\kappa_A, 1_A] \circ (\epsilon_A \vee 1_A) \circ d_{A,\neg A,A} \circ (1_A \wedge \eta_A) \circ \langle 1_A, !_A \rangle. \tag{2}$$

The defining zero-identity equation asserts that

$$f \circ 0_A = 0_C \circ g \qquad (f, g : A \to C). \tag{3}$$

Equation (2) is $(0\eta\epsilon)$ of [1, §14.2, p. 317]. Its product and coproduct unit isomorphisms are expressed here in pairing and copairing notation. Equation (3) is the zero-identity law $(0I)$; see [1, §14.3, p. 322].

Appendix B displays equations (1)–(3) as commutative diagrams.

Whenever $\mathsf{B}(A, C)$ is nonempty, define $0_{A,C} = f \circ 0_A$ for any $f : A \to C$. Equation (3) makes this independent of f: both $f \circ 0_A$ and $g \circ 0_A$ equal $0_C \circ g$. Taking $f = g$ gives $f \circ 0_A = 0_C \circ f$. Consequently zero arrows absorb composition whenever the composites are defined: $h \circ 0_{A,C} = 0_{A,D}$ for $h : C \to D$, and $0_{A,C} \circ k = 0_{X,C}$ for $k : X \to A$.

Two arrows are *parallel* when they have the same source and target. A *proof congruence* on B is an equivalence relation on each hom-set preserved by composition and every arrow operation of the presentation, including the derived operations of pairing, copairing, and negation. A *substitution congruence* is a proof congruence also preserved by every uniform substitution $\sigma : P \to \mathrm{Form}(P)$. Here substitution acts on the entire arrow term, including its source and target; this is rule (su) of [1, §2.7, p. 51].

The *total congruence* identifies every parallel pair. A category is *thin*, or a *preorder*, when each hom-set has at most one element. We call B *maximal* when its only substitution congruences are equality and the total congruence.

Theorem 2.1. *Let P be a nonempty set of propositional variables. The freely generated zero-identity Boolean category $\mathsf{B}(P)$ of Došen and Petrić is maximal. Equivalently, every substitution congruence on $\mathsf{B}(P)$ that identifies two distinct parallel arrows identifies all parallel arrows.*

Sections 4 and 5 give the proof.

3 Occurrences and substitution

For a formula A, write $\mathrm{Occ}(A)$ for the finite set of positions occupied by propositional variables in A. Repeated occurrences of the same variable occupy distinct positions. We use disjoint unions to distinguish these positions.

For finite sets X, Y, let $X \sqcup Y = (\{1\} \times X) \cup (\{2\} \times Y)$ be their tagged disjoint union. The occurrence sets are defined recursively by

$$\begin{aligned} \mathrm{Occ}(p) &= \{*\}, & \mathrm{Occ}(\top) &= \mathrm{Occ}(\bot) = \varnothing, \\ \mathrm{Occ}(\neg A) &= \mathrm{Occ}(A), & \mathrm{Occ}(A \star C) &= \mathrm{Occ}(A) \sqcup \mathrm{Occ}(C) \quad (\star \in \{\wedge, \vee\}). \end{aligned}$$

The label map $\lambda_A : \mathrm{Occ}(A) \to P$ assigns to each occurrence its underlying variable. Distinct positions may have the same label. Numbering the occurrences from left to right gives a bijection $\mathrm{Occ}(A) \to \{1, \ldots, |\mathrm{Occ}(A)|\}$; the latter set is empty when $|\mathrm{Occ}(A)| = 0$. Appendix A.2 gives the substitution formula in these finite-ordinal coordinates.

Let $\mathbf{Rel}_{\mathrm{fin}}$ be the category of finite sets and binary relations. A relation $R : X \to Y$ is a subset of $X \times Y$; identities are diagonal relations, and composition is relational composition. Transporting the occurrence interpretation of [1] to the tagged coordinates gives a functor $G : \mathsf{B} \to \mathbf{Rel}_{\mathrm{fin}}$ with $GA = \mathrm{Occ}(A)$. For $f : A \to C$, its image is the relation $Gf \subseteq \mathrm{Occ}(A) \times \mathrm{Occ}(C)$. For example,

$$G\pi_1^{A,C} = \{((1, x), x) : x \in \mathrm{Occ}(A)\}, \qquad G\iota_1^{A,C} = \{(x, (1, x)) : x \in \mathrm{Occ}(A)\}.$$

The second projection and injection use tag 2. Pairing takes the union of the two relations after tagging their targets; copairing takes the union after tagging their sources. The relation $Gd_{A,B,C}$ is the graph of the rebracketing bijection

$$\mathrm{Occ}(A) \sqcup (\mathrm{Occ}(B) \sqcup \mathrm{Occ}(C)) \longrightarrow (\mathrm{Occ}(A) \sqcup \mathrm{Occ}(B)) \sqcup \mathrm{Occ}(C),$$

which preserves each occurrence of A, B, C [1, §2.9, p. 60]. Zero arrows have empty relations.

Theorem 3.1 (Boolean Coherence, Došen–Petrić)**.** *The functor $G : \mathsf{B} \to \mathbf{Rel}_{\mathrm{fin}}$ is faithful: parallel arrows f, g of B are equal if and only if $Gf = Gg$.*

This is [1, §14.2, p. 321]. In particular, for $p \in P$, the two projections $p \wedge p \to p$ have different occurrence relations, so B is nonthin.

Joyal's collapse result says that a cartesian closed category with an initial object $\bot$ is a preorder if $A \cong (A \Rightarrow \bot) \Rightarrow \bot$ for every object A [1, §14.3, p. 324]. The category B is not cartesian closed. Faithfulness gives at most one arrow $\top \to E$ for every E, since $\mathrm{Occ}(\top)$ is empty, whereas $1_p \neq 0_p$ for $p \in P$. An exponential $p \Rightarrow p$ would give a bijection $\mathsf{B}(p, p) \cong \mathsf{B}(\top, p \Rightarrow p)$, a contradiction.

A *literal* is p or $\neg p$ for $p \in P$. A formula is in *negation normal form* when negation occurs only immediately before a propositional variable. The double-negation and De Morgan isomorphisms, including the isomorphisms for the truth constants, give an isomorphism $a_A : A \to A^\circ$ to a formula in negation normal form. The relation Ga_A is the graph of a bijection $\mathrm{Occ}(A) \to \mathrm{Occ}(A^\circ)$ [1, §14.1, pp. 313–315].

For arrows between formulas in negation normal form, every pair in the occurrence relation consists of occurrences of the same literal. In the presentation on negation normal forms in [1, §14.2, pp. 318–321], objects are formulas built from literals, conjunction, disjunction and the truth constants. Arrow terms are generated by identities, associativity and symmetry maps, diagonals and codiagonals, projections and injections, mix, dissociativity, unit maps, zero endomorphisms, and η_p, ϵ_p. The zero endomorphisms and η_p, ϵ_p have empty relations; each remaining generator joins equal literals. Composition, $\wedge$ and $\vee$ preserve this property. The equivalence with B preserves the occurrence interpretation, so the assertion follows.

Substitution replaces each occurrence of a variable by a separate block of occurrences of its substituted formula. Appendix A.1 describes these blocks and their label maps by commutative diagrams, following [11, §9]. Let Q be a set of propositional variables and $\sigma : P \to \mathrm{Form}(Q)$ a substitution. If $A[\sigma]$ denotes the result of uniform substitution, there is a bijection

$$\mathrm{Occ}(A[\sigma]) \cong D_{A,\sigma}, \qquad D_{A,\sigma} = \coprod_{x \in \mathrm{Occ}(A)} \mathrm{Occ}(\sigma(\lambda_A(x))). \tag{4}$$

The set on the right consists of pairs (x, t), where x is an occurrence of A and t is an occurrence within the formula inserted there. Induction on A proves the bijection: the variable case has one

block, the constant cases have none, binary connectives form tagged unions, and negation preserves the occurrence set. In particular, two occurrences of the same variable give two separately tagged copies of its substituted formula.

Put $E_\sigma = \coprod_{p \in P} \mathrm{Occ}(\sigma(p))$. Define $\mathrm{pr}_D : D_{A,\sigma} \to \mathrm{Occ}(A)$ and $\mathrm{pr}_E : E_\sigma \to P$ by $\mathrm{pr}_D(x,t) = x$ and $\mathrm{pr}_E(p,t) = p$. The function $\widehat{\lambda}_{A,\sigma} : D_{A,\sigma} \to E_\sigma$ sends (x,t) to $(\lambda_A(x), t)$. Thus

$$\begin{array}{ccc} D_{A,\sigma} & \xrightarrow{\widehat{\lambda}_{A,\sigma}} & E_\sigma \\ {\scriptstyle \mathrm{pr}_D}\downarrow & & \downarrow{\scriptstyle \mathrm{pr}_E} \\ \mathrm{Occ}(A) & \xrightarrow[\lambda_A]{} & P. \end{array} \tag{5}$$

Both composites send (x,t) to $\lambda_A(x)$. For $(x,t),(x',t) \in D_{A,\sigma}$, their images under $\widehat{\lambda}_{A,\sigma}$ agree exactly when $\lambda_A(x) = \lambda_A(x')$, even if $x \neq x'$.

The same presentation defines $\mathsf{B}(Q)$ when Q is empty. Substitution on arrow terms induces a functor

$$F_\sigma : \mathsf{B}(P) \longrightarrow \mathsf{B}(Q), \qquad A \longmapsto A[\sigma], \quad f \longmapsto f[\sigma].$$

Here $f : A \to C$ is sent to $f[\sigma] : A[\sigma] \to C[\sigma]$. Rule (su) makes this functor well defined on arrow terms modulo the defining equations. It preserves identities, composition, and the specified arrow operations. For each arrow, the blocks in (4) describe the occurrence sets of its substituted source and target.

4 Isolating an occurrence

Put $j_A = \iota_1^{A,\top} : A \to A \vee \top$. We shall derive the equation

$$j_A \circ \pi_1^{A,\bot} = 0_{A\wedge\bot, A\vee\top}. \tag{K_A}$$

The right-hand side is also $\iota_2^{A,\top} \circ !_\bot \circ \pi_2^{A,\bot}$: its occurrence relation is empty, so Boolean Coherence identifies it with the zero arrow. Thus (K_A) is the equation used in [1, §9.7, pp. 215–216].

Lemma 4.1 (Isolation of an occurrence)**.** *For distinct parallel arrows $f, g : A \to C$ in B, there are a literal ℓ and arrows*

$$i : \ell \wedge \bot \longrightarrow A, \qquad r : C \longrightarrow \ell \vee \top$$

such that, after possibly interchanging f and g,

$$r \circ f \circ i = j_\ell \circ \pi_1^{\ell,\bot}, \qquad r \circ g \circ i = 0_{\ell\wedge\bot, \ell\vee\top}.$$

Proof. First replace f, g by $a_C \circ f \circ a_A^{-1}$ and $a_C \circ g \circ a_A^{-1}$. These remain distinct, so it suffices to construct the arrows for the negation normal forms A°, C°. If $i^\circ : \ell \wedge \bot \to A^\circ$ and $r^\circ : C^\circ \to \ell \vee \top$ satisfy the required equations, then $i = a_A^{-1} \circ i^\circ : \ell \wedge \bot \to A$ and $r = r^\circ \circ a_C : C \to \ell \vee \top$ satisfy them for the original formulas. We may therefore assume that A, C are in negation normal form.

Faithfulness of G gives a pair belonging to exactly one of Gf, Gg. Interchange f, g if necessary, and choose $(x,y) \in Gf \setminus Gg$. The occurrences x and y have the same literal ℓ. Set $D = \ell \wedge \bot$ and $E = \ell \vee \top$. For every formula V, the arrows

$$b_V = \kappa_V \circ \pi_2^{\ell,\bot} : D \to V, \qquad t_V = \iota_2^{\ell,\top} \circ !_V : V \to E$$

have empty occurrence relations.

Construct $i : D \to A$ by induction on A, following the subformula that contains x at each binary connective. At the selected literal, use $\pi_1^{\ell,\bot}$. At a conjunction, pair the recursively constructed arrow with b_V for the unselected factor, placing it in the order specified by the tag. At a disjunction, compose the recursively constructed arrow with the injection of the selected summand. The displayed rules for G show inductively that Gi consists of one pair, whose first component is the sole occurrence of D and whose second component is x.

Construct $r : C \to E$ by induction on C, following the subformula that contains y. At the selected literal, use $\iota_1^{\ell,\top}$. At a conjunction, first project to the selected factor and then apply the recursively constructed arrow. At a disjunction, copair that arrow with t_V on the unselected summand. Thus Gr consists of one pair, whose first component is y and whose second component is the sole occurrence of E.

Relational composition now makes $G(r \circ f \circ i)$ the singleton relation from $\mathrm{Occ}(D)$ to $\mathrm{Occ}(E)$, whereas $G(r \circ g \circ i)$ is empty. Boolean Coherence gives the asserted equations. □

Proposition 4.2. *If a substitution congruence θ on* B *identifies distinct parallel arrows, then* (K_A) *holds modulo θ for every formula A.*

Proof. Lemma 4.1 and compatibility with composition give (K_A) for a literal ℓ. If $\ell = p$, let σ substitute A for p and fix the other variables. This yields (K_A) for A: under (4), the literal occurrence in each endpoint becomes the block $\mathrm{Occ}(A)$, and the arrow operations are preserved by F_σ; see also Appendix A.3.

If $\ell = \neg p$, substitute $\neg A$ for p, obtaining the equation for $\neg\neg A$. Precompose with $n_A^{-1} \wedge 1_\bot$ and postcompose with $n_A \vee 1_\top$, where $n_A : \neg\neg A \to A$. Put $k_A = j_A \circ \pi_1^{A,\bot} : A \wedge \bot \to A \vee \top$. The product and coproduct equations give the commutative square

$$\begin{array}{ccc} A \wedge \bot & \xrightarrow{\;k_A\;} & A \vee \top \\ {\scriptstyle n_A^{-1}\wedge 1_\bot}\downarrow & & \uparrow{\scriptstyle n_A \vee 1_\top} \\ \neg\neg A \wedge \bot & \xrightarrow[\;k_{\neg\neg A}\;]{} & \neg\neg A \vee \top. \end{array}$$

Precomposition and postcomposition of the zero arrow give a zero arrow by absorption. The equation for $\neg\neg A$ therefore gives (K_A) for A. □

5 Equality of the product projections

The next lemma identifies two isomorphisms of the types listed in [1, §12.5, p. 294].

Lemma 5.1. *For every formula A, the arrows*

$$q_A = [j_A \circ \pi_1^{A,\bot}, \iota_2^{A,\top}] : (A \wedge \bot) \vee \top \longrightarrow A \vee \top,$$
$$e_A = 1_A \wedge j_A : A \wedge A \longrightarrow A \wedge (A \vee \top)$$

are isomorphisms in B.

Proof. First take $A = p \in P$. By [1, §12.5, p. 294], the source and target of q_p are isomorphic. Each has one occurrence of p, so the relation of any such isomorphism is the unique bijection between these occurrence sets. The arrow q_p has the same relation. Boolean Coherence therefore identifies q_p with that isomorphism.

The same cited list gives an isomorphism $b : p \wedge p \to p \wedge (p \vee \top)$: take $n = m = 1$ in $p^n \wedge (p^m \vee \top) \cong p^{n+m}$ and choose $p^2 = p \wedge p$. The relation Gb is a bijection between two-element occurrence sets. After precomposing b with the product symmetry if necessary, it agrees with the order-preserving bijection Ge_p. Coherence then identifies e_p with an isomorphism. These two identifications also follow from the occurrence-bijection construction in the proof of [2, Lemma 7].

For arbitrary A, substitute A for p and fix the other variables. The functor F_σ sends q_p, e_p to q_A, e_A and preserves their inverse equations. □

Lemma 5.2. *Let θ be a proof congruence on B such that (K_A) holds modulo θ for every formula A. Then B/θ is a preorder.*

Proof. Write $\equiv$ for equality modulo θ. The quotient retains products and coproducts: their pairing, copairing, and defining equations pass to equivalence classes. Fix A, and put $E = A \vee \top$, $S = (A \wedge \bot) \vee \top$, and $s = \iota_2^{A,\top} : \top \to E$. The arrows $0_{A\wedge\bot,E}$ and $s \circ !_{A\wedge\bot}$ have empty occurrence relations and hence are equal in B. Thus (K_A) and the coproduct equations give

$$q_A \equiv [s \circ !_{A\wedge\bot}, s] = s \circ !_S.$$

The inverse equations for q_A hold in the quotient, so

$$1_E = q_A \circ q_A^{-1} \equiv s \circ !_S \circ q_A^{-1} = s \circ !_E.$$

Together with $!_E \circ s = 1_\top$, this makes E terminal in the quotient.

Consequently the projection $r = \pi_1^{A,E} : A \wedge E \to A$ is invertible, with inverse $\langle 1_A, s \circ !_A \rangle$. Lemma 5.1 makes e_A invertible too, and the product equations give $r \circ e_A = \pi_1^{A,A}$. Hence $\pi_1^{A,A}$ is invertible. Its right inverse is the diagonal $\Delta_A = \langle 1_A, 1_A \rangle : A \to A \wedge A$, which must therefore be its inverse. It follows that

$$\pi_2^{A,A} \equiv \pi_2^{A,A} \circ \Delta_A \circ \pi_1^{A,A} = \pi_1^{A,A}.$$

For parallel arrows $f, g : X \to A$, pairing gives $f = \pi_1^{A,A} \circ \langle f, g \rangle \equiv \pi_2^{A,A} \circ \langle f, g \rangle = g$. Thus the quotient is a preorder. □

Proof of Theorem 2.1. Suppose that a substitution congruence θ on $\mathsf{B}(P)$ identifies two distinct parallel arrows. Proposition 4.2 gives (K_A) modulo θ for every formula A. Lemma 5.2 then makes θ total. □

We also obtain maximality in the sense of [1, §9.3, p. 194]. An *interpretation* in a zero-identity Boolean category $\mathcal{C}$ assigns an object of $\mathcal{C}$ to each propositional variable. For an assignment $v : P \to \mathrm{Ob}(\mathcal{C})$, the induced functor $I_v : \mathsf{B}(P) \to \mathcal{C}$ preserves the specified products, coproducts, units, dissociativity arrows, negation maps, η and ϵ.

Corollary 5.3. *Let P be a nonempty set of propositional variables, and let $f, g : A \to C$ be distinct arrows of $\mathsf{B}(P)$. If a zero-identity Boolean category $\mathcal{C}$ satisfies $f = g$ under every interpretation of the propositional variables, then $\mathcal{C}$ is a preorder.*

Proof. Declare two parallel arrows of $\mathsf{B}(P)$ equivalent when all their interpretations in $\mathcal{C}$ agree. Since every I_v preserves the arrow operations, this relation is a proof congruence. For a substitution $\sigma : P \to \mathrm{Form}(P)$, the composite $I_v \circ F_\sigma$ is the interpretation induced by $p \mapsto I_v(\sigma(p))$. The congruence is therefore preserved by substitution. It contains $f = g$, so Theorem 2.1 makes it total. In particular, $\pi_1^{p,p} = \pi_2^{p,p}$ holds under every interpretation, for any $p \in P$. Interpret p as an arbitrary object X of $\mathcal{C}$. Its two projections $X \wedge X \to X$ are equal. For any arrows $a, b : Y \to X$, composing $\langle a, b \rangle : Y \to X \wedge X$ with the equal projections gives $a = b$. □

6 Lean formalization

In the Lean 4 development [12], formulas and arrow terms are inductive types, with arrow terms indexed by their source and target. The relation `Derives` is generated by the presentation's equations and congruence rules. Its quotient is the category $\mathsf{B}(P)$. Substitution, normalization and the occurrence interpretation are defined on terms and proved to respect `Derives`.

The theorem `relative_maximality` states that a substitution congruence identifying distinct parallel arrows is total. It assumes the arrow operations, a faithful occurrence interpretation, isomorphisms to negation normal forms, preservation of literals along occurrence relations, uniform substitution and double negation, with their equations. Its proof that the quotient is a preorder uses the arrow

$$h_A = [1_A, 1_A] \circ m_{A,A} : A \wedge A \longrightarrow A,$$

where the mix arrow $m_{A,C} : A \wedge C \to A \vee C$ is given, for any formula B, by

$$m_{A,C} = (\pi_1^{A,B} \vee 1_C) \circ d_{A,B,C} \circ (1_A \wedge \iota_2^{B,C}).$$

This is equation (dm) of [1, §11.1, p. 234]. The right-hand side is independent of B. The symmetry of mix gives $h_A \circ \langle \pi_2^{A,A}, \pi_1^{A,A} \rangle = h_A$. Write $\equiv$ for equality modulo a proof congruence containing (K_A) for every formula. For $B = \bot$, put

$$v_A = d_{A,\bot,A} \circ (1_A \wedge \iota_2^{\bot,A}) : A \wedge A \longrightarrow (A \wedge \bot) \vee A.$$

The coproduct equations and (K_A) give

$$\begin{aligned} j_A \circ h_A &= [j_A \circ \pi_1^{A,\bot}, j_A] \circ v_A \\ &\equiv [0_{A\wedge\bot, A\vee\top}, j_A] \circ v_A = j_A \circ \pi_2^{A,A}. \end{aligned}$$

The last equality holds in B by Boolean Coherence: both occurrence relations send the second copy of $\mathrm{Occ}(A)$ identically to the A-summand of $\mathrm{Occ}(A \vee \top)$ and have no pairs from the first copy. Precomposing with the symmetry of $A \wedge A$ and using the preceding identity for h_A gives $j_A \circ \pi_1^{A,A} \equiv j_A \circ \pi_2^{A,A}$. Precompose this equation with $\langle 1_A, 0_A \rangle$. Since the composite $j_A \circ 0_A$ has empty occurrence relation, coherence gives

$$j_A \equiv \iota_2^{A,\top} \circ !_A.$$

Put $s_A = \iota_2^{A,\top} : \top \to A \vee \top$. The coproduct equations give

$$1_{A\vee\top} = [j_A, s_A] \equiv [s_A \circ !_A, s_A] = s_A \circ !_{A\vee\top}.$$

Since $!_{A\vee\top} \circ s_A = 1_\top$, the object $A \vee \top$ is terminal in the quotient. Composing with the coproduct symmetry gives $\iota_2^{\top,A} \equiv \iota_1^{\top,A} \circ !_A$. For $B = \top$, put $w_A = 1_A \wedge !_A : A \wedge A \to A \wedge \top$. Substituting the preceding equation for $\iota_2^{\top,A}$ in the expression for mix gives

$$\begin{aligned} h_A &\equiv [\pi_1^{A,\top}, 1_A] \circ d_{A,\top,A} \circ (1_A \wedge \iota_1^{\top,A}) \circ w_A \\ &= [\pi_1^{A,\top}, 1_A] \circ \iota_1^{A\wedge\top,A} \circ w_A \\ &= \pi_1^{A,\top} \circ w_A = \pi_1^{A,A}. \end{aligned}$$

The second line uses (1); the third uses the coproduct and product equations. The symmetry of h_A yields $\pi_1^{A,A} \equiv \pi_2^{A,A}$, and pairing equates any two parallel arrows.

Cut elimination and Boolean Coherence are proved for the presentation defining $\mathsf{B}(P)$. Normalization, preservation of literals and uniform substitution on terms and their source and target formulas satisfy the remaining hypotheses of `relative_maximality`. Its application gives `Presentation.maximality`.

The theorem `Presentation.short_maximality` formalizes the proof in Section 5, with explicit inverses for q_A and e_A. The model corollary is proved using interpretations preserving the specified operations and their composition with substitution. The formalization also verifies the occurrence maps and finite-ordinal constructions of Appendix A, the diagram identities of Appendix B, and the cut reductions and rank comparisons discussed in Appendix C.

Five declarations are registered in the Palomar Registry [13]: the two maximality proofs, the congruence dichotomy, nonzero variable identities, and the absence of arrows from $\top$ to $\bot$.

A Substitution in occurrence coordinates

Došen and Petrić's generalized composition of definable connectives [1, §2.5, pp. 45–46] substitutes formula patterns and computes the map labelling the resulting occurrences. We describe this map by the tagged coordinates of [11, §§9.2–9.3]. Each occurrence after substitution is indexed by an occurrence of the outer pattern and an occurrence of the pattern inserted there.

A.1 Occurrence maps and their fibres

For an integer $n \geq 0$, put $\underline{n} = \{1, \ldots, n\}$, with $\underline{0} = \varnothing$. A *formula pattern* on a finite set of placeholders is a formula in those placeholders. Its *occurrence map* records the index of the placeholder at each position, after enumerating the placeholders and reading positions from left to right.

Let $U = \{p_1, \ldots, p_k\}$ consist of distinct placeholders and let $M \in \mathrm{Form}(U)$ have m occurrences. Its occurrence map is

$$o_M : \underline{m} \longrightarrow \underline{k}, \qquad o_M(r) = i \quad \text{when position } r \text{ contains } p_i.$$

For each $i \in \underline{k}$, choose a set $Q_i = \{q_{i,1}, \ldots, q_{i,\ell_i}\}$ of distinct local placeholders and a pattern $N_i \in \mathrm{Form}(Q_i)$ with n_i occurrences. The sets Q_i are pairwise disjoint. Write $o_i : \underline{n_i} \to \underline{\ell_i}$ for the occurrence map of N_i. Empty placeholder sets and empty occurrence sets are allowed; a pattern over an empty placeholder set has no variable occurrences.

Define the sets of occurrence coordinates and placeholder indices by

$$D = \{(r, t) : r \in \underline{m},\ t \in \underline{n_{o_M(r)}}\},$$
$$C = \{(i, s) : i \in \underline{k},\ s \in \underline{\ell_i}\}.$$

Thus (r, t) specifies an occurrence in the copy of $N_{o_M(r)}$ inserted at position r of M. The pair (i, s) specifies a local placeholder of N_i. Define the injections

$$a_r : \underline{n_{o_M(r)}} \longrightarrow D, \quad a_r(t) = (r, t), \qquad b_i : \underline{\ell_i} \longrightarrow C, \quad b_i(s) = (i, s).$$

There is a unique function $\gamma : D \to C$ for which, for every $r \in \underline{m}$, the following square commutes:

$$\begin{array}{ccc} \underline{n_{o_M(r)}} & \xrightarrow{a_r} & D \\ {\scriptstyle o_{o_M(r)}}\downarrow & & \downarrow{\scriptstyle \gamma} \\ \underline{\ell_{o_M(r)}} & \xrightarrow[b_{o_M(r)}]{} & C. \end{array} \tag{6}$$

Every element of D is $a_r(t)$ for a unique pair (r,t). Thus these squares determine γ uniquely by

$$\gamma(r,t) = \big(o_M(r), o_{o_M(r)}(t)\big).$$

Define the projections $p_D : D \to \underline{m}$ and $p_C : C \to \underline{k}$ by $p_D(r,t) = r$ and $p_C(i,s) = i$. Then $p_C \circ \gamma = o_M \circ p_D$, so the following square commutes:

$$\begin{array}{ccc} D & \xrightarrow{\ \gamma\ } & C \\ {\scriptstyle p_D}\downarrow & & \downarrow{\scriptstyle p_C} \\ \underline{m} & \xrightarrow[\ o_M\ ]{} & \underline{k}. \end{array} \tag{7}$$

Both composites send (r,t) to $o_M(r)$. If $r \neq r'$ but $o_M(r) = o_M(r') = i$, the copies of $\underline{n_i}$ indexed by r and r' are disjoint subsets of D.

To identify this construction with pattern substitution, replace each $q_{i,s}$ in N_i by (i,s) and denote the resulting pattern over C by $\widetilde{N}_i$. Let $\Phi_0 : U \to \mathrm{Form}(C)$ be given by $\Phi_0(p_i) = \widetilde{N}_i$. Every occurrence of $M[\Phi_0]$ has unique coordinates $(r,t) \in D$. This gives a bijection $\chi : \mathrm{Occ}(M[\Phi_0]) \to D$, and its relation to the label map is the commutative square

$$\begin{array}{ccc} \mathrm{Occ}(M[\Phi_0]) & \xrightarrow[\ \cong\ ]{\ \chi\ } & D \\ {\scriptstyle \lambda_{M[\Phi_0]}}\downarrow & & \downarrow{\scriptstyle \gamma} \\ C & \xrightarrow[\ 1_C\ ]{} & C. \end{array} \tag{8}$$

The occurrence with coordinates (r,t) contains the placeholder $\big(o_M(r), o_{o_M(r)}(t)\big)$.

A.2 The finite-ordinal presentation

The finite ordinals used in [1] number the elements of each block consecutively. Put

$$s_0 = t_0 = 0, \qquad s_r = \sum_{q=1}^{r} n_{o_M(q)} \quad (1 \le r \le m), \qquad t_i = \sum_{j=1}^{i} \ell_j \quad (1 \le i \le k).$$

Numbering each disjoint union consecutively gives bijections

$$\begin{aligned} \theta_D : D &\longrightarrow \underline{s_m}, \quad \theta_D(r,t) = s_{r-1} + t, \\ \theta_C : C &\longrightarrow \underline{t_k}, \quad \theta_C(i,s) = t_{i-1} + s. \end{aligned}$$

Define $\delta : \underline{s_m} \to \underline{t_k}$ as the unique function making the following square commute:

$$\begin{array}{ccc} D & \xrightarrow{\ \gamma\ } & C \\ {\scriptstyle \cong}\downarrow{\scriptstyle \theta_D} & & \cong\downarrow{\scriptstyle \theta_C} \\ \underline{s_m} & \xrightarrow[\ \delta\ ]{} & \underline{t_k}. \end{array} \tag{9}$$

Equivalently, $\delta = \theta_C \circ \gamma \circ \theta_D^{-1}$. For each $(r,t) \in D$, this gives

$$\delta(s_{r-1} + t) = t_{o_M(r)-1} + o_{o_M(r)}(t) \qquad \big((r,t) \in D\big). \tag{10}$$

Every source position belongs to exactly one nonempty block, so this formula determines δ even when some blocks are empty.

In Došen and Petrić's notation, μ and ν_i correspond to o_M and o_i, and $\mu(\nu_1, \ldots, \nu_k)$ corresponds to δ. For $x \in \underline{s_m}$, let r be the unique index with $s_{r-1} < x \leq s_r$. Substituting $t = x - s_{r-1}$ in (10) gives their formula [1, §2.5, p. 46]. The square therefore describes the function on finite ordinals in their generalized composition. The *skeleton* of a pattern is obtained by replacing every placeholder by one fixed symbol and leaving the constants and connectives unchanged. A skeleton and an occurrence map recover the pattern by assigning the indicated placeholder to each position. The generalized composite is therefore determined by the skeleton of $M[\Phi_0]$ and the map δ.

For example, take

$$M(p, q) = q \wedge (p \vee q), \qquad N_1(a, b) = a \wedge b, \qquad N_2(c, d) = d \wedge (c \vee d).$$

The occurrence maps, listed by successive values, are $o_M = (2, 1, 2)$, $o_1 = (1, 2)$, and $o_2 = (2, 1, 2)$. The source blocks have sizes $3, 2, 3$, while the target blocks list (a, b) and (c, d). Thus

source position	1	2	3	4	5	6	7	8
placeholder	d	c	d	a	b	d	c	d
δ	4	3	4	1	2	4	3	4

For instance, source position 2 has coordinates $(1, 2)$. The function γ gives $\gamma(1, 2) = (2, 1)$, and $\theta_C(2, 1) = 3$. Thus $\delta(2) = 3$.

A.3 Shared variables and the maximality argument

The local placeholders may be assigned names in a common variable set Q. Let $\rho : C \to Q$ be that assignment; different elements of C may have the same image. Replace each placeholder $c \in C$ by the variable $\rho(c)$, obtaining the renaming map $(-)[\rho] : \mathrm{Form}(C) \to \mathrm{Form}(Q)$. Define $\sigma : U \to \mathrm{Form}(Q)$ by $\sigma(p_i) = \widetilde{N}_i[\rho]$. Then the following square commutes:

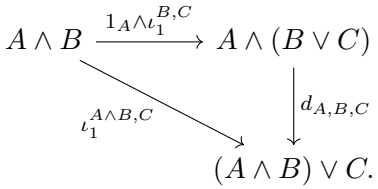
$$\begin{array}{ccc} \mathrm{Form}(U) & \xrightarrow{(-)[\Phi_0]} & \mathrm{Form}(C) \\ {\scriptstyle (-)[\sigma]}\downarrow & & \downarrow{\scriptstyle (-)[\rho]} \\ \mathrm{Form}(Q) & \xrightarrow[1]{} & \mathrm{Form}(Q). \end{array} \tag{11}$$

At p_i, both routes give $\sigma(p_i)$. The constant and connective clauses agree, so structural induction on the formula proves commutativity. Renaming changes the labels on the occurrences. Thus the same variable of Q may occur in several inserted patterns while the pairs (r, t) distinguish its occurrences.

If the literal isolated in Proposition 4.2 is p, equation (K_A) has source $p \wedge \bot$ and target $p \vee \top$. Each contains one occurrence of p. Substituting A for p therefore gives one block $\mathrm{Occ}(A)$ at each endpoint, and the substituted arrows have type $A \wedge \bot \to A \vee \top$. If the isolated literal is $\neg p$, the substitution $p \mapsto \neg A$ instead gives $\neg\neg A \wedge \bot \to \neg\neg A \vee \top$; the double-negation square in Proposition 4.2 then gives the required equation for A.

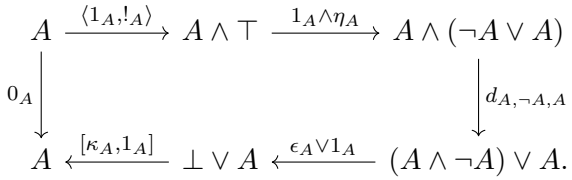
B Diagrams for three defining equations

The first-injection equation. For formulas A, B, C, equation (1) is the commutative triangle



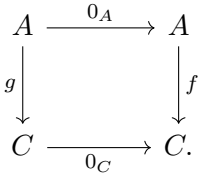
Both paths have source $A \wedge B$ and target $(A \wedge B) \vee C$. The upper arrow is induced by $1_A : A \to A$ and $\iota_1^{B,C} : B \to B \vee C$.

The zero identity. For a formula A, equation (2) is expressed by the commutative diagram



The path across the top, down the right side, and back along the bottom is the five-arrow composite in (2). Its first arrow pairs $1_A : A \to A$ with $!_A : A \to \top$; its last arrow copairs $\kappa_A : \bot \to A$ with $1_A : A \to A$. The arrows between them use $\eta_A : \top \to \neg A \vee A$ and $\epsilon_A : A \wedge \neg A \to \bot$, together with the dissociativity arrow.

The zero law. For any parallel arrows $f, g : A \to C$, equation (3) is the commutative square



The upper-right path is $f \circ 0_A : A \to C$, and the lower-left path is $0_C \circ g : A \to C$. The arrows f and g need not be equal.

C Two corrections to the cut-elimination proof

The cut-elimination proof in [1, §11.2] contains two typographical errors in its rank comparisons. We use the notation of that proof: r is the rank of the cut being reduced and r' is the rank of a recursive subcut. The left and right ranks are positive; their sum is the cut rank [1, p. 249].

Page 260, case $(w\,\gamma\,3)$**.** In the sentence immediately following the displayed reduction, the inequality $r < r'$ should read $r' < r$. The recursive cuts have the same cut formula and smaller rank: the corresponding one-sided ranks decrease on passing to the indicated premises, while the other one-sided rank is unchanged.

Page 262, case $(w \vee 1)$**.** In the continuation of this case, the phrase "whose rank is 1" should read "whose right rank is 1", as in the analogous argument on p. 261. The selected formula in the right premise is newly introduced by the displayed $\vee L$ rule, so its right rank is 1. Its total rank is the unchanged left rank plus 1, and is therefore at least 2.

Use of AI tools

OpenAI's ChatGPT assisted with the development of the proof, preparation of the manuscript, and checks of the arguments and references. OpenAI Codex was used to develop the Lean formalization and to run its verification checks. The author takes responsibility for the mathematical claims and the final text.

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